Xinglang Zhang

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Education

Huazhong University of Science and Technology (HUST) Sept 2024 – Jun 2028

B.S. in Cyber Science and Engineering

- **GPA:** 4.80/5.0 **Weighted Avg:** 94.51 **Rank:** **1/108**
- **Selected Coursework:** Calculus A (I/II) (A+), Linear Algebra (A+), Discrete Math (A+), Data Structures (A+), Assembly Language (A+), Probability Theory (A+), Digital Circuit (A+), Database Systems (A+).
- **Research Interests:** LLM Reasoning, LLM Test-Time Training, Neural-Symbolic AI, Agents, Tool Learning

Publications

Logical Phase Transitions: Understanding Collapse in LLM Logical Reasoning
Xinglang Zhang, Yunyao Zhang*, et al.* | **ACL 2026 (Main Conference)** | [ArXiv] | [Code]

Semantic-Aware Logical Reasoning via a Semiotic Framework
*Yunyao Zhang, **Xinglang Zhang**, et al.* | **ACL 2026 (Oral)** | [ArXiv] | [Code]

IntervenSim: Intervention-Aware Social Network Simulation for Opinion Dynamics
*Yunyao Zhang, Zuocheng Ying, **Xinglang Zhang*** | [ArXiv] | [Code]

Too Good to Be Real: Synthetic Preference Misreads What Real Humans Reward
***Xinglang Zhang**, Yuanmeng Xiang, et al.* | **Conference Submission**

Semiotic Logical Hexagon Theory for LLM Logical Reasoning
Yunyao Zhang, **Xinglang Zhang***, et al.* | [ArXiv] | [Code] | **Journal Submission**

Social Intelligence Modeling: A Comprehensive Survey from Social Perception to Social Simulation
*Zikai Song, Xiajie Li, Yunyao Zhang, **Xinglang Zhang**, et al.* | [Paper] | [GitHub]

EarthVerse: Benchmarking Scientific Agents Across Dynamic Earth Systems and Natural Hazards
*Zhiqing Cui, Xinxiang Yin, Yihong Tang, **Xinglang Zhang**, et al.* | [Project Page] | [Paper]

Research Experience

State-Aware Test-Time Training for Tool-Using Language Models Jul 2026 – Present

Research Intern, BLENDER Lab, University of Illinois Urbana-Champaign (UIUC)
Mentor: Cheng Qian **Advisor:** Heng Ji
Research Areas: Test-Time Training, Tool Learning, LLM Agents, Adaptive Data Selection

- **Framework:** Formulated tool-learning adaptation as a state-aware test-time training problem—deciding which self-generated experiences to learn, at what dose, in which order, and when to stop—and introduced *Transfer Value* to quantify their marginal utility on future queries.
- **Method:** Developed a label-free hierarchical experience-selection framework that combines cluster-level NLL probes, confidence signals, and structural diversity, avoiding exhaustive Source–Target transfer evaluation.
- **Findings:** Demonstrated that experience utility depends on the current Solver state, exposure dose, and training order; identified a transition from early tool-protocol learning to later semantic learning, with 25/32 held-out experiences flipping from beneficial to harmful after adaptation.
- **Results:** Without access to any external ground-truth or target labels, improved Exact accuracy from 7.71% for the base model to 38.93% (+31.22 pp). State-aware reselection further outperformed static selection by +8.95 pp in Exact accuracy. Preparing an **ICLR** submission.

Hexagonal Semantic Reasoning for Reliable LLMs Mar 2026 – Jul 2026

Research Assistant, Intelligent Media Computing and Network Security Lab, HUST
Advisor: Zikai Song, Research Areas: Neuro-Symbolic AI, Semantic Reasoning, LLM Reliability

- **Framework:** Proposed *HexLogicAgent*, a semantic-aware logical reasoning framework that organizes proposition meanings through a logical hexagon before deduction and verification.
- **Theory:** Developed a formal logical hexagon theory for semantic reasoning.
- **Results:** Achieved consistent gains across five logical reasoning benchmarks.
- **Outcome:** Co-first author paper submitted to **Artificial Intelligence**.

Diagnosing and Correcting Synthetic-Real Preference Misalignment in Online Argumentation Jan 2026 – May 2026

Project Leader, Intelligent Media Computing and Network Security Lab, HUST
Advisor: Zikai Song, Research Areas: Preference Alignment, Social Media Analysis, Controllable Text Generation

- **Framework:** Proposed *Ontological Preference Measurement* to diagnose synthetic-real preference mismatch across logic, affect, and expression dimensions.

- **Method:** Developed *OMRA* to mask and reconstruct reasoning scaffolds while preserving stance and coherence.
- **Results:** Reduced marginal, geometric, and transfer gaps by 54.4% on average across four LLM families; preparing a first-author **EMNLP 2026** submission.

Exploring Logical Reasoning Boundaries and Enhancement Strategies in LLMs via Neural-Symbolic Methods — Aug 2025 – Jan 2026

Project Leader, Intelligent Media Computing and Network Security Lab, HUST
Advisor: Zikai Song, Research Areas: LLM Evaluation, Logical Reasoning, Neural-Symbolic AI

- **Framework:** Identified *Logical Phase Transitions* in LLM reasoning and proposed a *Neural-Symbolic Tuning* framework to stabilize reasoning under increasing logical complexity.
- **Results:** Across five benchmarks, the method mitigates reasoning collapse, yielding +1.26 (naive) and +3.95 (CoT) average accuracy gains with improved generalization.
- **Outcome:** Published as first author in **ACL 2026 Main Conference**.

Reasoning Framework for Large LLMs based on Philosophical Paradigms — Feb 2025 – Aug 2025

Research Assistant, Intelligent Media Computing and Network Security Lab, HUST
Advisor: Zikai Song, Research Areas: LLM Reasoning, Neural-Symbolic AI, Semantic and Symbolic Reasoning

- **Framework & Benchmark:** Proposed *LogicAgent*, a semiotic-aware logical reasoning framework, and introduced *RepublicQA*, a benchmark jointly modeling logical and semantic complexity.
- **Results:** Achieved state-of-the-art results on RepublicQA (+**6.25%**) and strong generalization across four logical reasoning benchmarks (+**7.05%**).
- **Outcome:** Co-authored paper published in **ACL 2026 Main Conference**.

Honors

- **Outstanding Student**, Qiming College *(Highest Honor for HUST Undergraduates; Top 1.85%)* — 2025 – 2027
- **Jin Yugang Lan-Ying Scholarship** *(Top 1%)* — 2025 – 2026
- **National Scholarship** *(Highest Honor for Chinese Undergraduates; Top 0.3% Nationwide)* — 2024 – 2025
- **Triple-A ("Sanhao") Student Scholarship** *(Top 5.5%)* — 2024 – 2025
- **Outstanding Youth League Member** *(Top 4.1%)* — 2024 – 2025
- **Outstanding Youth League Cadre** *(Top 4.1%)* — 2024 – 2025
- **Academic Excellence Scholarship** & **Self-improvement Scholarship** *(Top 10%)* — 2024 – 2025

Service

ACL ARR 2026 May Reviewer

Leadership Experience

- **Class Representative, Qiming Experimental Class 2401** — Sept 2024 – Present
  - Organized academic events, managed administrative tasks, and mentored students to enhance class cohesion.
- **Head of Organization Department, School of Cyber Science and Engineering** — Sept 2025 – Present
  - Managed administrative and organizational affairs for the Communist Youth League.
- **Head of Lesson Planning, Peer Lecturer Group** — Apr 2025 – Present
  - Fostered academic excellence across the college through peer mentoring and curriculum planning.

Skills

ML & Research Stack: Python, C/C++, PyTorch, HuggingFace, LoRA, SFT, RLHF, vLLM, Poster Creation

Mathematical Foundations: Linear Algebra, Probability Theory, Calculus, Discrete Mathematics

Languages: English (TOEFL 5.5/6; W6 R6 L5.5 S5; CET-4: 609; CET-6: 611), Chinese (Native)